



ClinicalPrep AI — Functional PRD

Product Requirements, Personas, User Flows & Clinical Guardrails (v1.0 – v4.0+)

1. Executive Vision & Problem Statement

During 10-to-15 minute outpatient clinic visits, patients often forget 50% of critical symptom details due to anxiety or time pressure. Physicians spend valuable consultation time taking down basic history instead of diagnosing.

ClinicalPrep AI acts as an empathetic, conversational pre-visit assistant that interviews patients before their appointment and compiles their medical history into a 30-second readable **Doctor Brief** and downloadable PDF summary.

2. Target User Personas

Persona A: Elderly / Low Tech-Literacy Patient (e.g., Ramesh, 64)

- **Needs:** Cannot type easily on mobile keyboards; speaks in regional Hindi/Bengali dialect; needs auto-playing voice audio and large simple buttons.
- **Solution Fit:** Full voice input/output, WhatsApp voice note integration, auto-read TTS, large pain scale visual cards.

Persona B: Caregiver Son/Daughter (e.g., Ananya, 32)

- **Needs:** Completing appointment preparation on behalf of an aging parent while traveling or in the clinic waiting room.
- **Solution Fit:** Caregiver Mode (3rd person intake prompts), instant PDF generation, direct WhatsApp summary forwarding.

Persona C: Busy Outpatient Physician (e.g., Dr. Sharma, MD)

- **Needs:** Requires a structured 30-second summary highlighting Chief Complaint, HPI, Meds, and Red Flags before entering the exam room.
- **Solution Fit:** Standardized Doctor Brief PDF, zero-fluff bullet formatting, SMART-on-FHIR JSON export for EMR software.

3. Functional Module Requirements Matrix

Module ID	Feature Name	Functional Behavior & User Experience	Release
F-101	Slot-Filling Intake	Accepts unconstrained user paragraphs or audio transcripts, extracts all provided demographics and symptoms in 1 turn, and dynamically asks only for unfilled slots.	v2.0 (Must)
F-102	Voice Mode	1-tap microphone button captures patient voice, transcribes via Whisper, and auto-plays assistant responses using streaming TTS voice output.	v2.0 (Must)
F-103	Emergency Triage	Immediate detection of emergency keywords (chest pain, numbness, shortness of breath) halts intake and displays prominent 911/112 alert banners.	v2.0 (Must)
F-104	Token Cost Tracker	Real-time admin drawer displaying prompt tokens, completion tokens, and dollar cost for developer observability.	v2.0 (Must)
F-201	WhatsApp Channel	Allows patients to complete intake by exchanging voice notes with a WhatsApp Business Bot, receiving PDF summary directly in chat.	v3.0 (Good)
F-202	Local Idiom Engine	Translates local spoken expressions ("pet mein mitha mitha dard") into standardized clinical English ("mild dull abdominal ache").	v3.0 (Good)
F-301	Visual Body Map	Interactive 2D human silhouette where patients tap pain locations to auto-fill anatomical terms into the Doctor Brief.	v4.0 (Nice)
F-302	Post-Visit Decoder	Upload consultation audio or prescription photos to receive a simplified plain-language home care action plan.	v4.0 (Nice)

4. Clinical Guardrails & Safety Requirements

- 1. **Zero-Diagnosis Guarantee:** System prompt strict rules forbid offering diagnoses, speculative medical conditions, or drug dosage recommendations.
- 2. **Red-Flag Emergency Halting:** Instant triage takes precedence over intake completion.
- 3. **Prominent Medical Disclaimer:** Displayed on initial chat screen and hardcoded into PDF footer: *"ClinicalPrep AI is a preparation tool and does not provide medical advice or diagnosis."*